

Time-Series Analysis

Statistics and Data Analysis

Unit-6

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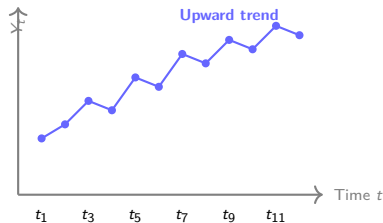


- 1 Time-Series Fundamentals
 - Stationarity and Differencing
- 2 Autocorrelation
- 3 Smoothing Techniques
 - Simple Moving Average
 - Exponential Smoothing
- 4 Autoregressive (AR) and Moving Average (MA) Models
- 5 Forecasting Fundamentals
- 6 ARIMA Models

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A **time series** is an ordered sequence of observations recorded at successive, equally spaced points in time. Each observation is indexed by the time at which it was recorded.

- The ordering of data is **critical** — unlike cross-sectional data, shuffling destroys information.
- Denoted Y_t for $t = 1, 2, \dots, T$ where T is the total number of observations.
- Observations may be recorded at any regular frequency: hourly, daily, weekly, monthly, quarterly, or yearly.



Real-World Examples

Daily stock prices, monthly airline passengers, hourly temperature readings, quarterly GDP, annual population figures, weekly retail sales.

Any time series can be decomposed into fundamental building blocks. The **classical decomposition** model expresses this as:

Additive: $Y_t = T_t + S_t + C_t + I_t$

Multiplicative: $Y_t = T_t \times S_t \times C_t \times I_t$

Trend (T_t)

Long-term
direction

Seasonal (S_t)

Fixed-period
repeating pattern

Cyclical (C_t)

Long waves,
no fixed period

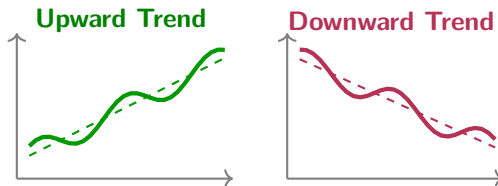
Irregular (I_t)

Random noise,
unpredictable

Additive vs. Multiplicative

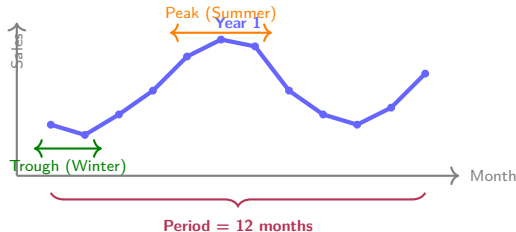
Use **additive** when seasonal fluctuations are roughly constant in magnitude. Use **multiplicative** when fluctuations grow proportionally with the level of the series.

Trend (T_t) represents the underlying **long-term increase, decrease, or stagnation** in a time series, abstracted from seasonal and irregular effects.



Trend Type	Description	Example
Linear upward	Steady constant increase	Annual GDP growth
Linear downward	Steady constant decrease	Print newspaper sales
Non-linear (exponential)	Accelerating growth	Internet users (2000s)
No trend (stationary)	Fluctuates around a constant mean	Daily temperature deviations

Seasonality (S_t) refers to regular, predictable fluctuations that recur over a **known, fixed period** (e.g. 12 months, 7 days, 24 hours).



Common Seasonal Patterns

Yearly: ice cream sales, heating bills

Weekly: restaurant traffic (weekends peak)

Daily: website traffic (peaks at noon)

Seasonality $\neq$ Cyclical

Seasonality has a **fixed, known** period. Cyclical patterns (e.g. business cycles) have **variable, unknown** periods, typically longer than a year.

Numerical Example — Additive Decomposition

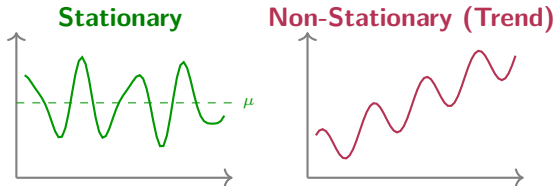
Consider the following quarterly sales data (in thousands) over 3 years. We separate each observation into its components.

Year	Qtr	Actual (Y_t)	Trend (T_t)	Seasonal (S_t)	Residual (I_t)	Check: $T + S + I$
	Q1	120	150	-35	+5	$150 - 35 + 5 = \mathbf{120}$ ✓
	Q2	180	155	+20	+5	$155 + 20 + 5 = \mathbf{180}$ ✓
	Q3	210	160	+50	0	$160 + 50 + 0 = \mathbf{210}$ ✓
	Q4	130	165	-35	0	$165 - 35 + 0 = \mathbf{130}$ ✓
	Q1	140	170	-35	+5	$170 - 35 + 5 = \mathbf{140}$ ✓
	Q2	195	175	+20	0	$175 + 20 + 0 = \mathbf{195}$ ✓
	Q3	230	180	+50	0	$180 + 50 + 0 = \mathbf{230}$ ✓
	Q4	150	185	-35	0	$185 - 35 + 0 = \mathbf{150}$ ✓

Observations

The **trend** increases by roughly 5 units per quarter. The **seasonal** pattern repeats identically: Q1 and Q4 are below trend (-35), Q2 is above (+20), and Q3 peaks (+50). The **residuals** are small and random.

A time series is **stationary** if its statistical properties — mean, variance, and autocorrelation structure — remain **constant over time**.



Formally, a **strictly stationary** process satisfies: the joint distribution of $(Y_{t_1}, Y_{t_2}, \dots, Y_{t_k})$ is identical to $(Y_{t_1+h}, Y_{t_2+h}, \dots, Y_{t_k+h})$ for all h and all choices of time points.

Why Stationarity Matters

Most classical time series models (AR, MA, ARMA, ARIMA) **require stationarity**. A non-stationary series must first be transformed — most commonly by **differencing**.

Differencing removes trend by computing the change between consecutive observations:

$$Y'_t = Y_t - Y_{t-1}.$$

Numerical Example: First Differencing

t	1	2	3	4	5	6	7	8
Y_t	100	108	119	126	138	145	159	168
$Y'_t = Y_t - Y_{t-1}$	—	8	11	7	12	7	14	9

The original series has a strong upward trend. The differenced series (Y'_t) fluctuates around $\bar{Y}' = 9.7$ with no obvious trend — it appears **stationary**.

Second Differencing ($d = 2$)

If first differencing does not remove the trend, apply it again: $Y''_t = Y'_t - Y'_{t-1} = (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2})$
 $= Y_t - 2Y_{t-1} + Y_{t-2}$

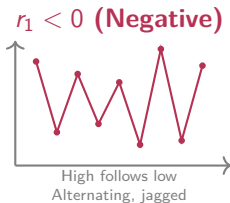
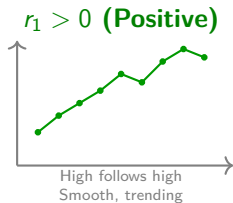
Practical Rule

In practice, $d = 1$ suffices for most time series. Rarely is $d > 2$ needed. **Over-differencing** introduces artificial patterns and should be avoided.

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Autocorrelation measures the **linear relationship between a time series and a lagged version of itself**. It quantifies the “memory” of the series.

$$r_k = \frac{\sum_{t=k+1}^T (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^T (Y_t - \bar{Y})^2} \quad r_k \in [-1, +1]$$



Interpretation

$r_k \approx +1$: strong positive correlation at lag k . $r_k \approx 0$: no linear relationship. $r_k \approx -1$: strong negative correlation at lag k .

Compute the **lag-1 autocorrelation** r_1 for the series: $Y = \{2, 5, 4, 8, 7, 10\}$.

Step 1: Compute the mean. $\bar{Y} = (2 + 5 + 4 + 8 + 7 + 10)/6 = 36/6 = 6.0$

Step 2: Form the paired table (original vs. lag-1).

t	Y_t	Y_{t-1}	$(Y_t - \bar{Y})$	$(Y_{t-1} - \bar{Y})$	Product
2	5	2	-1	-4	$(-1)(-4) = +4$
3	4	5	-2	-1	$(-2)(-1) = +2$
4	8	4	+2	-2	$(+2)(-2) = -4$
5	7	8	+1	+2	$(+1)(+2) = +2$
6	10	7	+4	+1	$(+4)(+1) = +4$
Sum of Products (Numerator)					+8

Step 3: Compute the denominator $= \sum_{t=1}^6 (Y_t - \bar{Y})^2 = 16 + 1 + 4 + 4 + 1 + 16 = 42$.

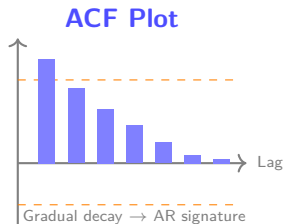
Step 4: $r_1 = 8/42 = 0.190$ (weak positive autocorrelation at lag 1).

Interpretation

A value of 0.19 suggests mild positive dependence between consecutive observations — knowing today's value gives a slight hint about tomorrow's.

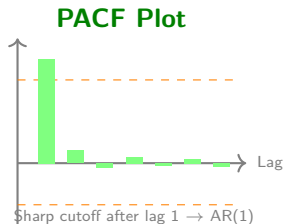
ACF (Autocorrelation Function)

- Plots r_k for $k = 1, 2, 3, \dots$
- Shows the **total** correlation at each lag, including indirect effects through intermediate lags.
- Dashed lines at $\pm 1.96/\sqrt{T}$ denote 95% significance bounds.



PACF (Partial ACF)

- Plots partial autocorrelation at lag k after **removing** the effects of lags 1 through $k-1$.
- Measures the **direct** relationship between Y_t and Y_{t-k} .
- Critical for identifying AR order p .



ACF/PACF Identification Rules — Summary Table

The patterns observed in ACF and PACF plots provide the primary diagnostic tool for selecting the appropriate model.

ACF Behaviour	PACF Behaviour	Suggested Model	Interpretation
Dies out gradually (exponential or sinusoidal decay)	Cuts off sharply after lag p	AR(p)	Underdifferenced; add AR terms
Cuts off sharply after lag q	Dies out gradually	MA(q)	Overdifferenced; add MA terms
Dies out gradually	Dies out gradually	ARMA(p, q)	Mixed; try small p and q
All lags insignificant	All lags insignificant	White Noise	No autocorrelation; no model needed

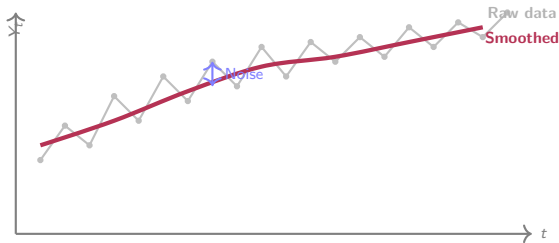
Key Rules of Thumb

Positive lag-1 autocorrelation typically suggests an AR signature (series slightly underdifferenced). **Negative** lag-1 autocorrelation suggests an MA signature (series slightly overdifferenced).

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Why Smoothing? — Separating Signal from Noise

Raw time series data is often contaminated by **random fluctuations (noise)** that obscure the true underlying pattern (signal). Smoothing techniques **average out the noise** to reveal trends and level.



Moving Average

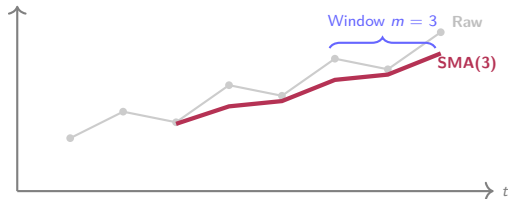
Uses a sliding window of m observations; all observations weighted **equally**.

Exponential Smoothing

Assigns **exponentially decreasing** weights to older observations; more responsive.

The **m -period simple moving average** replaces each observation with the unweighted mean of the most recent m observations:

$$SMA_t = \frac{Y_t + Y_{t-1} + Y_{t-2} + \dots + Y_{t-m+1}}{m} = \frac{1}{m} \sum_{i=0}^{m-1} Y_{t-i}$$



Effect of Window Size m

Small m : Responsive but noisy.

Large m : Smooth but lags behind.

Average lag = $(m + 1)/2$ periods.

Limitations

- All m observations weighted equally
- Completely ignores older data
- Cannot extrapolate trends

Compute the **3-period** and **5-period moving average** for the following monthly sales data.

t	Month	Y_t	3-Period SMA	Calculation (3-SMA)	5-Period SMA
1	Jan	200	—	—	—
2	Feb	220	—	—	—
3	Mar	250	223.3	$(200+220+250)/3$	—
4	Apr	270	246.7	$(220+250+270)/3$	—
5	May	300	273.3	$(250+270+300)/3$	248.0
6	Jun	280	283.3	$(270+300+280)/3$	264.0
7	Jul	260	280.0	$(300+280+260)/3$	272.0
8	Aug	290	276.7	$(280+260+290)/3$	280.0
9	Sep	320	290.0	$(260+290+320)/3$	290.0
10	Oct	340	316.7	$(290+320+340)/3$	298.0

Forecast for Next Period

3-SMA forecast for $t = 11$: $(320+340+Y'_{10})/3$ — uses the last 3 values.

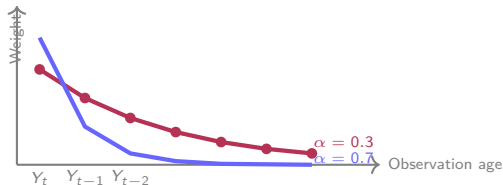
5-SMA forecast for $t = 11$: $(260+290+320+340+Y'_{10})/5$ — uses the last 5 values.

The 5-SMA is **smoother** but **slower to react** to the recent upturn.

SES addresses the limitation of SMA by assigning **exponentially decreasing weights** to older observations. Recent data receives more importance.

$$F_{t+1} = \alpha \cdot Y_t + (1 - \alpha) \cdot F_t \quad 0 \leq \alpha \leq 1$$

Equivalently: $F_{t+1} = F_t + \alpha(Y_t - F_t) = F_t + \alpha \cdot e_t$ (error-correction form)



Key Property: Average Age of Data

The average age of data in SES forecasts is $1/\alpha$. An SES model with $\alpha = 0.2$ has an average age of 5 periods, comparable to a 9-period SMA.

Given sales data $Y = \{200, 220, 250, 270, 300, 280, 260, 290\}$ and $\alpha = 0.3$. Let $F_1 = Y_1 = 200$.

t	Y_t	F_t	Calculation: $F_{t+1} = 0.3Y_t + 0.7F_t$	Error $e_t = Y_t - F_t$	e_t^2
1	200	200.0	Initial: $F_1 = Y_1$	0.0	0.0
2	220	200.0	$F_2 = 0.3(220) + 0.7(200) = \mathbf{206.0}$	+20.0	400.0
3	250	206.0	$F_3 = 0.3(250) + 0.7(206) = \mathbf{219.2}$	+44.0	1936.0
4	270	219.2	$F_4 = 0.3(270) + 0.7(219.2) = \mathbf{234.4}$	+50.8	2580.6
5	300	234.4	$F_5 = 0.3(300) + 0.7(234.4) = \mathbf{254.1}$	+65.6	4303.4
6	280	254.1	$F_6 = 0.3(280) + 0.7(254.1) = \mathbf{261.9}$	+25.9	670.8
7	260	261.9	$F_7 = 0.3(260) + 0.7(261.9) = \mathbf{261.3}$	-1.9	3.6
8	290	261.3	$F_8 = 0.3(290) + 0.7(261.3) = \mathbf{269.9}$	+28.7	823.7

$$\text{MSE} = \frac{400+1936+2580.6+4303.4+670.8+3.6+823.7}{7} = \frac{10718.1}{7} = \mathbf{1531.2}$$

$$\text{RMSE} = \sqrt{1531.2} = \mathbf{39.1}$$

Forecast for period 9: $F_9 = 269.9$

Effect of α

With $\alpha = 0.3$ the forecast lags behind the uptrend. A larger α (e.g. 0.7) would react faster but be noisier.

α Value	Behaviour	Best When	Equivalent To
$\alpha \rightarrow 0$ (e.g. 0.1)	Very smooth, heavy lag; long memory	Stable series, little noise	Long moving average
$\alpha \approx 0.3$	Moderate smoothing	General-purpose default	6-period SMA
$\alpha \rightarrow 1$ (e.g. 0.9)	Very responsive, noisy	Rapidly changing level	Naïve forecast
$\alpha = 1$	$F_{t+1} = Y_t$ exactly	—	Random walk
$\alpha = 0$	$F_{t+1} = F_1$ (never updates)	—	Constant (mean) model

Optimal α Selection

Choose the value of α that minimises the sum of squared one-step-ahead forecast errors (SSE):

$$SSE = \sum_{t=1}^T (Y_t - F_t)^2$$

Connection to ARIMA

SES is mathematically equivalent to an **ARIMA(0,1,1)** model without constant, where the MA(1) coefficient $\theta_1 = 1 - \alpha$.

Aspect	Simple Moving Average	Exponential Smoothing
Weighting	Equal weight to last m	Exponentially decreasing
Older data	Completely ignored (beyond m)	Never fully forgotten
Parameters	Window size m (integer)	Smoothing constant α (continuous)
Optimisation	Try different m , pick best	Solver minimises SSE for α
Responsiveness	Depends on m	Depends on α
Average age	$(m+1)/2$	$1/\alpha$
Long-term forecast	Flat (horizontal)	Flat (horizontal)

Practical Guideline

For the same average data age, SES generally outperforms SMA because it assigns **relatively more weight to the most recent observation**. SES is also easier to optimise since α is a continuous parameter.

Moving Average and Exponential Smoothing

```
import pandas as pd
import numpy as np
from statsmodels.tsa.holtwinters import (
    SimpleExpSmoothing)
# Simple Moving Average (window = 3)
df['SMA_3'] = df['Sales'].rolling(window=3).mean()
# Simple Moving Average (window = 5)
df['SMA_5'] = df['Sales'].rolling(window=5).mean()
# Simple Exponential Smoothing
ses = SimpleExpSmoothing(df['Sales'])
ses_fit = ses.fit(smoothing_level=0.3, optimized=
    False)
df['SES'] = ses_fit.fittedvalues
# Auto-optimised alpha
ses_opt = ses.fit(optimized=True)
print(f"Optimal alpha: " f"{ses_opt.params['
    smoothing_level']:.4f}")
```

Note

rolling(window=m) computes the SMA. Set optimized=True to let the solver find the best α automatically by minimising SSE.

Tip

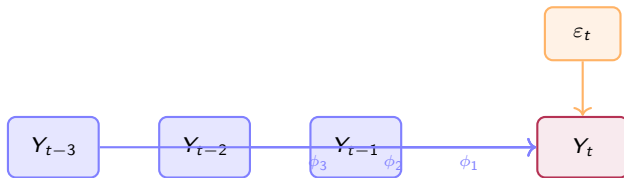
For series with **trend**, use Holt's method (ExponentialSmoothing with trend='add'). For **trend + seasonality**, use Holt-Winters.

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An **autoregressive model of order p** , denoted $AR(p)$, predicts the current value as a **linear combination of its own past values** plus a white noise error term.

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t$$

where c is a constant, $\phi_1, \dots, \phi_p$ are the AR coefficients, and $\varepsilon_t \sim WN(0, \sigma^2)$.



Stationarity Condition

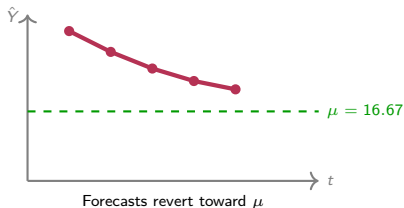
For $AR(1)$: $|\phi_1| < 1$. If $\phi_1 = 1$, the series becomes a **random walk** (non-stationary). The long-run mean of a stationary $AR(1)$ is $\mu = c/(1 - \phi_1)$.

Suppose we have estimated the model: $Y_t = 5 + 0.7 Y_{t-1} + \varepsilon_t$

Long-run mean: $\mu = c/(1 - \phi_1) = 5/(1 - 0.7) = 5/0.3 = \mathbf{16.67}$

If the current observation is $Y_{10} = 30$, forecast the next 5 periods:

Period	Calculation	$\hat{Y}_t$	Distance from μ
$t = 11$	$5 + 0.7 \times 30$	26.00	9.33
$t = 12$	$5 + 0.7 \times 26$	23.20	6.53
$t = 13$	$5 + 0.7 \times 23.2$	21.24	4.57
$t = 14$	$5 + 0.7 \times 21.24$	19.87	3.20
$t = 15$	$5 + 0.7 \times 19.87$	18.91	2.24



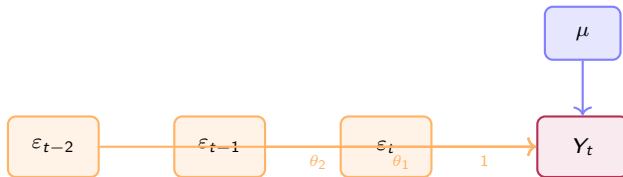
Mean Reversion

Each forecast moves 70% of the remaining distance toward $\mu = 16.67$. This **mean-reverting** behaviour is the hallmark of stationary AR models.

An **MA model of order q** , denoted $MA(q)$, expresses the current value as a linear combination of **current and past white noise error terms**.

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}$$

where μ is the mean, $\theta_1, \dots, \theta_q$ are MA coefficients, and $\varepsilon_t \sim WN(0, \sigma^2)$.



Critical Difference from AR

MA models have **limited memory**: an $MA(q)$ model only “remembers” the last q shocks. Forecasts revert to μ after q steps. AR models revert **gradually**.

Model: $Y_t = 50 + \varepsilon_t + 0.6\varepsilon_{t-1}$, $\mu = 50$, $\theta_1 = 0.6$.

Given the past errors below, compute Y_t and forecast future values.

t	ε_t	Y_t	Calculation
1	+3	53.0	$50 + 3 + 0.6(0) = 53.0$ (assume $\varepsilon_0 = 0$)
2	-2	49.8	$50 + (-2) + 0.6(3) = 50 - 2 + 1.8$
3	+4	52.8	$50 + 4 + 0.6(-2) = 50 + 4 - 1.2$
4	-1	51.4	$50 + (-1) + 0.6(4) = 50 - 1 + 2.4$
5	+5	54.4	$50 + 5 + 0.6(-1) = 50 + 5 - 0.6$

Forecasting from $t = 5$ (where $\varepsilon_5 = +5$):

Horizon	Calculation	$\hat{Y}$
$\hat{Y}_6$	$50 + 0 + 0.6(5) = 50 + 3$	53.0
$\hat{Y}_7$	$50 + 0 + 0.6(0) = 50$	50.0
$\hat{Y}_8$	$50 + 0 + 0 = 50$	50.0

Key Observation

After just **one step** ahead, the MA(1) forecast snaps back to the mean $\mu = 50$. The effect of the last shock $\varepsilon_5 = 5$ persists for exactly one forecast period.

Property	AR(p)	MA(q)
Predicts using	Past values $Y_{t-1}, \dots, Y_{t-p}$	Past errors $\varepsilon_{t-1}, \dots, \varepsilon_{t-q}$
ACF pattern	Tails off gradually	Cuts off after lag q
PACF pattern	Cuts off after lag p	Tails off gradually
Memory	Long (gradual mean reversion)	Short (snaps to μ after q steps)
Stationarity	Requires $ \phi < 1$ conditions	Always stationary (finite q)
Invertibility	Always invertible (finite p)	Requires $ \theta < 1$ conditions
Equivalent ARIMA	ARIMA($p, 0, 0$)	ARIMA($0, 0, q$)
Estimation	OLS (linear regression on lags)	Non-linear optimisation

Duality Principle

A stationary AR(p) process can be expressed as an MA(∞), and an invertible MA(q) process can be written as an AR(∞). This **duality** is why ARMA models combine both.

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Forecasting is the process of making **predictions about future values** of a time series based on historical data and an identified model.

Step 1: Collect and visualise data



Step 2: Stationarise (differencing, transforms)



Step 3: Identify model (ACF/PACF analysis)



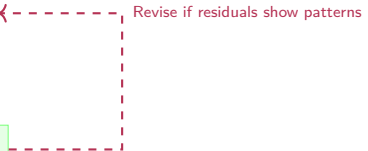
Step 4: Estimate parameters



Step 5: Diagnostic checking (residual analysis)



Step 6: Produce forecasts and confidence intervals



Evaluating how well our model predicts requires comparing forecasts $\hat{Y}_t$ to actual values Y_t .

t	Y_t	$\hat{Y}_t$	$e_t = Y_t - \hat{Y}_t$	$ e_t $	e_t^2	$ e_t/Y_t \times 100$
1	100	95	+5	5	25	5.00%
2	110	112	-2	2	4	1.82%
3	105	108	-3	3	9	2.86%
4	120	115	+5	5	25	4.17%
5	130	128	+2	2	4	1.54%
Sums:			+7	17	67	15.39%

ME = $7/5 = +1.4$ (slight positive bias)

MAE = $17/5 = 3.4$

MSE = $67/5 = 13.4$

RMSE = $\sqrt{13.4} = 3.66$

MAPE = $15.39/5 = 3.08\%$

Which Metric to Use?

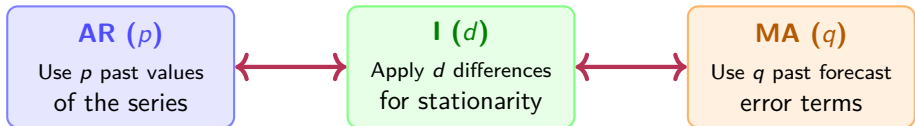
RMSE: penalises large errors heavily; same units as data. **MAE:** robust to outliers. **MAPE:** scale-independent, good for comparing across series. **ME:** detects systematic bias (positive = under-forecasting).

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 - Stationarity and Differencing
- 2 Autocorrelation
- 3 Smoothing Techniques
 - Simple Moving Average
 - Exponential Smoothing
- 4 Autoregressive (AR) and Moving Average (MA) Models
- 5 Forecasting Fundamentals
- 6 ARIMA Models**

ARIMA (**A**uto**R**egressive **I**ntegrated **M**oving **A**verage) unifies AR, differencing, and MA into a single powerful framework.

$$\text{ARIMA}(p, d, q) : Y'_t = c + \sum_{i=1}^p \phi_i Y'_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}$$

where Y'_t denotes the series after d differences.



Why “Integrated”?

A non-stationary series that can be made stationary by differencing d times is said to be **integrated of order d** , written $I(d)$. The forecast equation is then applied to the differenced series, and results are “integrated back” (cumulated) to obtain forecasts on the original scale.

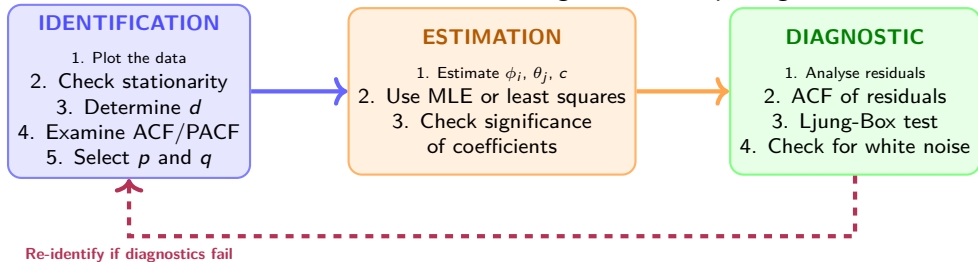
Every classical time series model is a **special case** of the general $ARIMA(p, d, q)$ family.

Model	p	d	q	ARIMA Notation
White noise	0	0	0	ARIMA(0,0,0)
Random walk	0	1	0	ARIMA(0,1,0) no constant
Random walk with drift	0	1	0	ARIMA(0,1,0) with constant
AR(1)	1	0	0	ARIMA(1,0,0)
AR(2)	2	0	0	ARIMA(2,0,0)
MA(1)	0	0	1	ARIMA(0,0,1)
Simple Exponential Smoothing	0	1	1	ARIMA(0,1,1) no constant
Holt's Linear Smoothing	0	2	2	ARIMA(0,2,2) no constant
Damped Trend Smoothing	1	1	2	ARIMA(1,1,2)

Elegance of ARIMA

By choosing different combinations of p , d , and q , one can represent naïve methods, exponential smoothing models, and sophisticated mixed AR-MA models within a single unified notation.

The **Box-Jenkins method** (1970) provides a systematic, iterative procedure for building ARIMA models. It consists of three stages, often requiring iteration.



Iterative Nature

If the diagnostic checks reveal patterns in the residuals (significant autocorrelation, non-constant variance), the analyst returns to the identification stage to revise the model. This loop continues until a satisfactory model is found.

Data: 12 months of product sales: $\{50, 55, 53, 60, 58, 65, 63, 70, 68, 75, 73, 80\}$

Step 1: Visual inspection. The data shows a clear **upward trend** $\Rightarrow$ non-stationary.

Step 2: First differencing ($d = 1$).

t	1	2	3	4	5	6	7	8	9	10	11	12
Y_t	50	55	53	60	58	65	63	70	68	75	73	80
Y'_t	—	5	-2	7	-2	7	-2	7	-2	7	-2	7

The differenced series alternates between -2 and 7 , with mean $\bar{Y}' = 30/11 = 2.73$. No apparent trend in $Y'_t \Rightarrow d = 1$ is sufficient.

Step 3: ACF/PACF of Y'_t .

The differenced series shows a strong **negative spike at lag 1** in the ACF that cuts off, and the PACF tails off. This is the **MA signature**.

Conclusion: Model Identification

ACF cuts off at lag 1 with negative spike $\Rightarrow$ **MA(1)** pattern. Therefore, we tentatively select **ARIMA(0, 1, 1)** with constant (drift).

Step 4: Parameter estimation (via software).

Suppose MLE yields: $\theta_1 = 0.80$, constant (drift) $c = 2.73$.

The forecasting equation becomes:

$$Y_t = Y_{t-1} + 2.73 + \varepsilon_t + 0.80 \varepsilon_{t-1}$$

Step 5: Forecasting from $t = 12$ where $Y_{12} = 80$ and suppose $\varepsilon_{12} = -1.5$:

Period	Calculation	$\hat{Y}_t$
$t = 13$	$80 + 2.73 + 0 + 0.80 \times (-1.5) = 80 + 2.73 - 1.20$	81.53
$t = 14$	$81.53 + 2.73 + 0 + 0.80 \times 0 = 81.53 + 2.73$	84.26
$t = 15$	$84.26 + 2.73 + 0 + 0 = 84.26 + 2.73$	86.99
$t = 16$	$86.99 + 2.73$	89.72

Key Observations

After the **first forecast**, the MA(1) effect vanishes (future errors assumed zero). From $t = 14$ onward, the model simply adds the drift $c = 2.73$ each period, producing a **linear trend** in long-term forecasts.

Suppose a different dataset, after first differencing ($d = 1$), shows: PACF cuts off at lag 1 $\Rightarrow$ AR(1) signature.

Estimated model: $(Y_t - Y_{t-1}) = 3 + 0.6(Y_{t-1} - Y_{t-2}) + \varepsilon_t$

Given: $Y_{12} = 200$, $Y_{11} = 190$.

Period	$\hat{Y}'_t$ (diff)	Calculation	$\hat{Y}_t$
$t = 13$	9.0	$3 + 0.6 \times (200 - 190) = 3 + 6$	$200 + 9.0 = \mathbf{209.0}$
$t = 14$	8.4	$3 + 0.6 \times 9.0 = 3 + 5.4$	$209.0 + 8.4 = \mathbf{217.4}$
$t = 15$	8.04	$3 + 0.6 \times 8.4 = 3 + 5.04$	$217.4 + 8.04 = \mathbf{225.4}$
$t = 16$	7.82	$3 + 0.6 \times 8.04 = 3 + 4.82$	$225.4 + 7.82 = \mathbf{233.3}$
...	$\rightarrow 7.5$	$c/(1 - \phi_1) = 3/0.4$	Steady growth

Long-Run Growth Rate

The period-to-period growth $\hat{Y}'_t$ converges to $c/(1 - \phi_1) = 3/0.4 = \mathbf{7.5}$ units per period. Unlike the ARIMA(0,1,1), the AR(1) component causes the growth rate itself to exhibit **gradual mean reversion**, starting at 9.0 and settling toward 7.5.

Residual Diagnostics — Validating Your ARIMA Model

After fitting an ARIMA model, the **residuals** $\hat{\varepsilon}_t = Y_t - \hat{Y}_t$ must resemble **white noise**. If they don't, the model has failed to capture all the information in the data.

1. ACF of residuals: no significant spikes

No remaining autocorrelation

2. Mean of residuals ≈ 0

No systematic bias

3. Constant variance (homoscedasticity)

No heteroscedasticity

4. Ljung-Box test: $p\text{-value} > 0.05$

Fail to reject H_0 : no autocorrelation

5. Residuals approximately normally distributed

Valid confidence intervals

If Diagnostics Fail

Return to the **identification** stage. Consider adding or removing AR/MA terms, changing d , or investigating seasonal patterns.

Building ARIMA Models in Python

```
import pandas as pd
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.graphics.tsaplots import (plot_acf,
                                           plot_pacf)

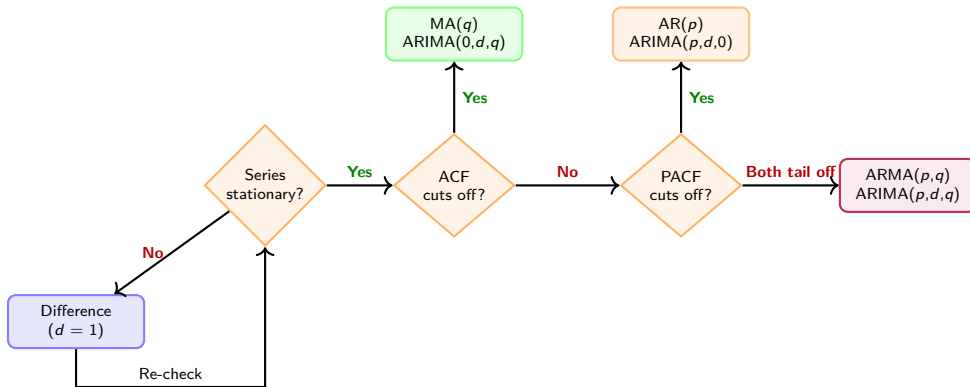
import pmdarima as pm # pip install pmdarima
# Step 1: Plot ACF and PACF
plot_acf(df['Sales'].diff().dropna(), lags=20)
plot_pacf(df['Sales'].diff().dropna(), lags=20)
# Step 2: Fit manual ARIMA(1,1,1)
model = ARIMA(df['Sales'], order=(1,1,1))
result = model.fit()
print(result.summary())
# Step 3: Forecast next 5 periods
forecast = result.forecast(steps=5)
# Auto-ARIMA: finds best (p,d,q)
auto = pm.auto_arima(df['Sales'], seasonal=False,
                    stepwise=True, trace=True)
print(auto.summary())
```

Auto-ARIMA

`pm.auto_arima()` automates the Box-Jenkins process: it tests multiple (p, d, q) combinations and selects the model with the lowest AIC/BIC.

AIC/BIC

AIC and **BIC** penalise model complexity. **Lower is better**. BIC penalises more heavily, favouring simpler models.



Parsimony Principle

Prefer simpler models. Avoid fitting both AR and MA terms unless clearly needed. In practice, models with $p + q \leq 2$ cover the vast majority of real-world time series.

Model/Technique	Key Formula	Parameters	Best For
Simple Moving Avg	$\frac{1}{m} \sum_{i=0}^{m-1} Y_{t-i}$	Window m	Smoothing, no trend
Exponential Smoothing	$\alpha Y_t + (1-\alpha)F_t$	$\alpha \in [0, 1]$	Level estimation
AR(p)	$c + \sum \phi_i Y_{t-i} + \varepsilon_t$	p, ϕ_i, c	Self-dependent data
MA(q)	$\mu + \varepsilon_t + \sum \theta_j \varepsilon_{t-j}$	q, θ_j, μ	Shock-driven data
ARIMA(p, d, q)	AR + Differencing + MA	p, d, q	General non-seasonal

Box-Jenkins Workflow: Identify → Estimate → Diagnose → Forecast

Problem 1 (Moving Average): Given data $Y = \{12, 18, 15, 21, 24, 19, 27\}$, compute the **3-period SMA** for $t = 3$ through $t = 7$. What is the forecast for $t = 8$?

Problem 2 (Exponential Smoothing): For $Y = \{80, 90, 85, 95, 100\}$ with $\alpha = 0.4$ and $F_1 = 80$, compute all forecasts F_2 through F_6 . Calculate the MAE.

Problem 3 (AR Forecasting): Model: $Y_t = 10 + 0.5 Y_{t-1} + \varepsilon_t$. If $Y_{10} = 30$, forecast $\hat{Y}_{11}$, $\hat{Y}_{12}$, $\hat{Y}_{13}$. What is the long-run mean? Verify forecasts converge toward it.

Problem 4 (Differencing): Given $Y = \{200, 210, 225, 245, 270\}$, compute the first differences and second differences. Is first differencing sufficient for stationarity?

Problem 5 (ARIMA Identification): A differenced series has ACF: lag 1 = 0.65, lag 2 = 0.3, lag 3 ≈ 0 ; PACF: lag 1 = 0.65, lag 2 = -0.2, lag 3 = -0.15, decaying gradually. What ARIMA model is appropriate?

Problem 6 (ARIMA Forecasting): An ARIMA(0,1,1) model has $\theta_1 = 0.5$ and drift $c = 4$. Given $Y_{20} = 150$ and $\varepsilon_{20} = -3$, forecast $\hat{Y}_{21}$, $\hat{Y}_{22}$, $\hat{Y}_{23}$.

Solution 1: $SMA_3 = 15.0, 18.0, 20.0, 21.3, 23.3$. Forecast for $t = 8$: $(24+19+27)/3 = \mathbf{23.3}$.

Solution 2: $F_2 = 80$, $F_3 = 0.4(90) + 0.6(80) = 84$, $F_4 = 0.4(85) + 0.6(84) = 84.4$, $F_5 = 0.4(95) + 0.6(84.4) = 88.6$, $F_6 = 0.4(100) + 0.6(88.6) = \mathbf{93.2}$. $MAE = (0 + 6 + 1 + 10.6 + 11.4)/5 = \mathbf{5.8}$.

Solution 3: $\hat{Y}_{11} = 10 + 0.5(30) = 25$, $\hat{Y}_{12} = 10 + 0.5(25) = 22.5$, $\hat{Y}_{13} = 10 + 0.5(22.5) = 21.25$. Long-run mean $= 10/(1 - 0.5) = \mathbf{20}$. Forecasts: $25 \rightarrow 22.5 \rightarrow 21.25 \rightarrow \dots \rightarrow 20 \checkmark$

Solution 4: First diffs: 10, 15, 20, 25 — still increasing (trend). Second diffs: 5, 5, 5 — constant $\Rightarrow$ stationary. First differencing alone is **not sufficient**; $d = 2$ is needed.

Solution 5: ACF cuts off after lag 2, PACF tails off $\Rightarrow$ **MA(2)** signature $\Rightarrow$ **ARIMA(0, 1, 2)**.

Solution 6: $\hat{Y}_{21} = 150 + 4 + 0.5(-3) = 152.5$. $\hat{Y}_{22} = 152.5 + 4 + 0 = 156.5$. $\hat{Y}_{23} = 156.5 + 4 = \mathbf{160.5}$.

Concepts

- $Y_t = T_t + S_t + C_t + I_t$ (additive decomposition)
- Stationarity: constant mean, variance, and autocorrelation
- Differencing (d times) removes trend
- ACF/PACF are the diagnostic keys for model identification

Models & Techniques

- SMA: equal weights, window m
- SES: $F_{t+1} = \alpha Y_t + (1-\alpha)F_t$
- AR(p): regression on own past
- MA(q): regression on past errors
- ARIMA(p, d, q): unifies all above

Core Takeaway

Time series analysis follows a principled workflow: **decompose** the series, ensure **stationarity**, identify the model via **ACF/PACF**, estimate and **diagnose**, then **forecast**. ARIMA provides a flexible, unified framework where smoothing methods, AR, and MA models are all special cases.

Thank You!

